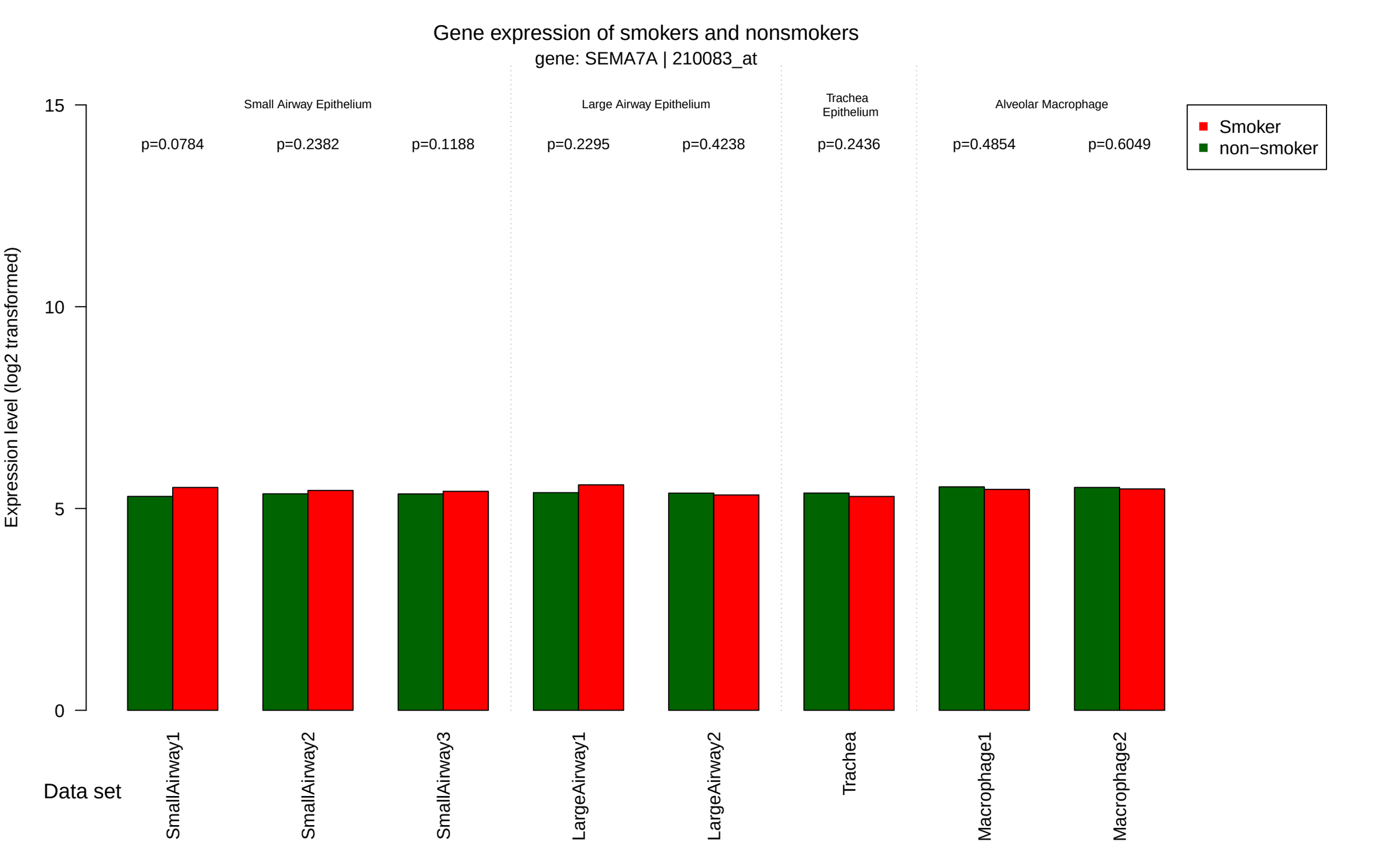
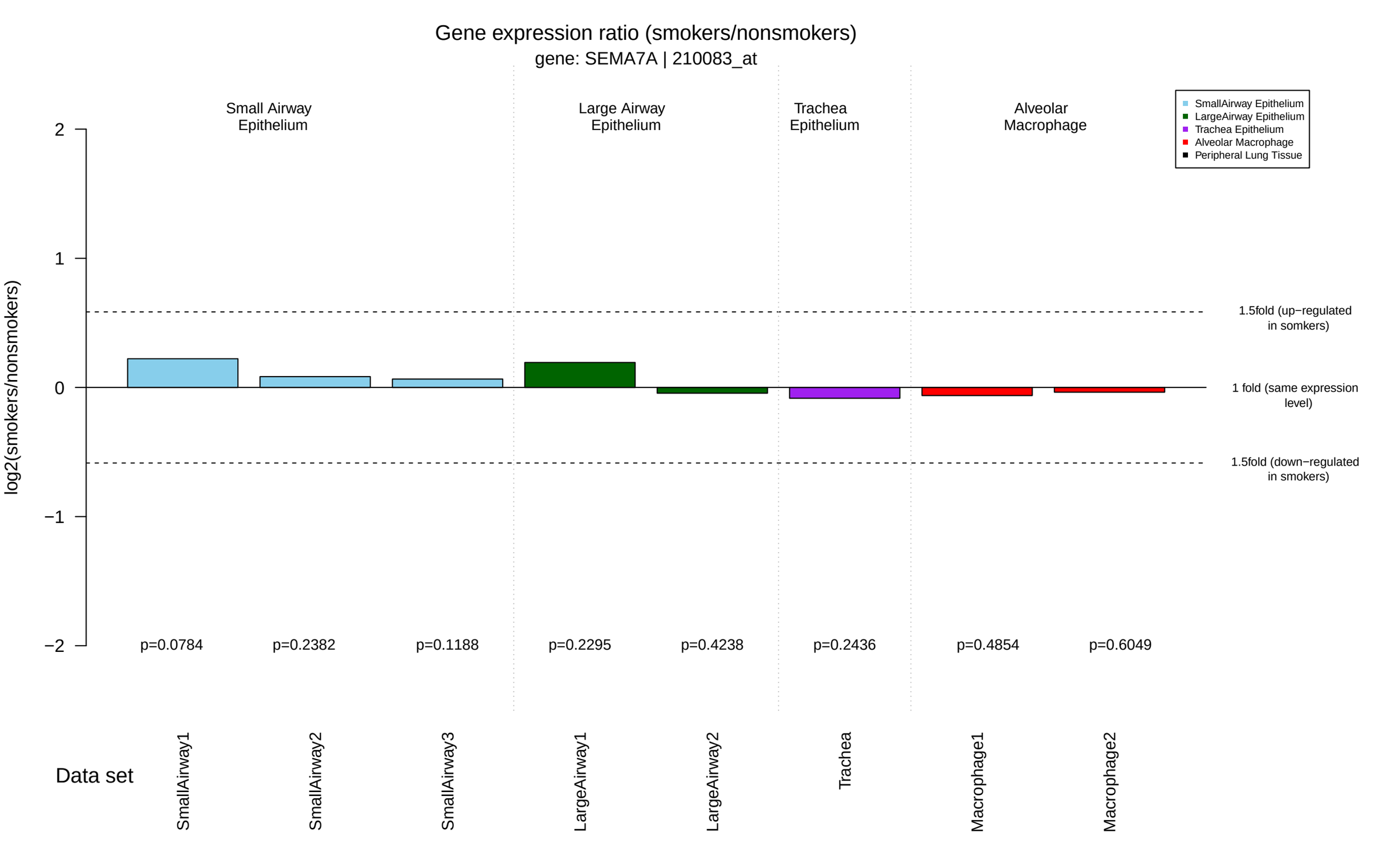
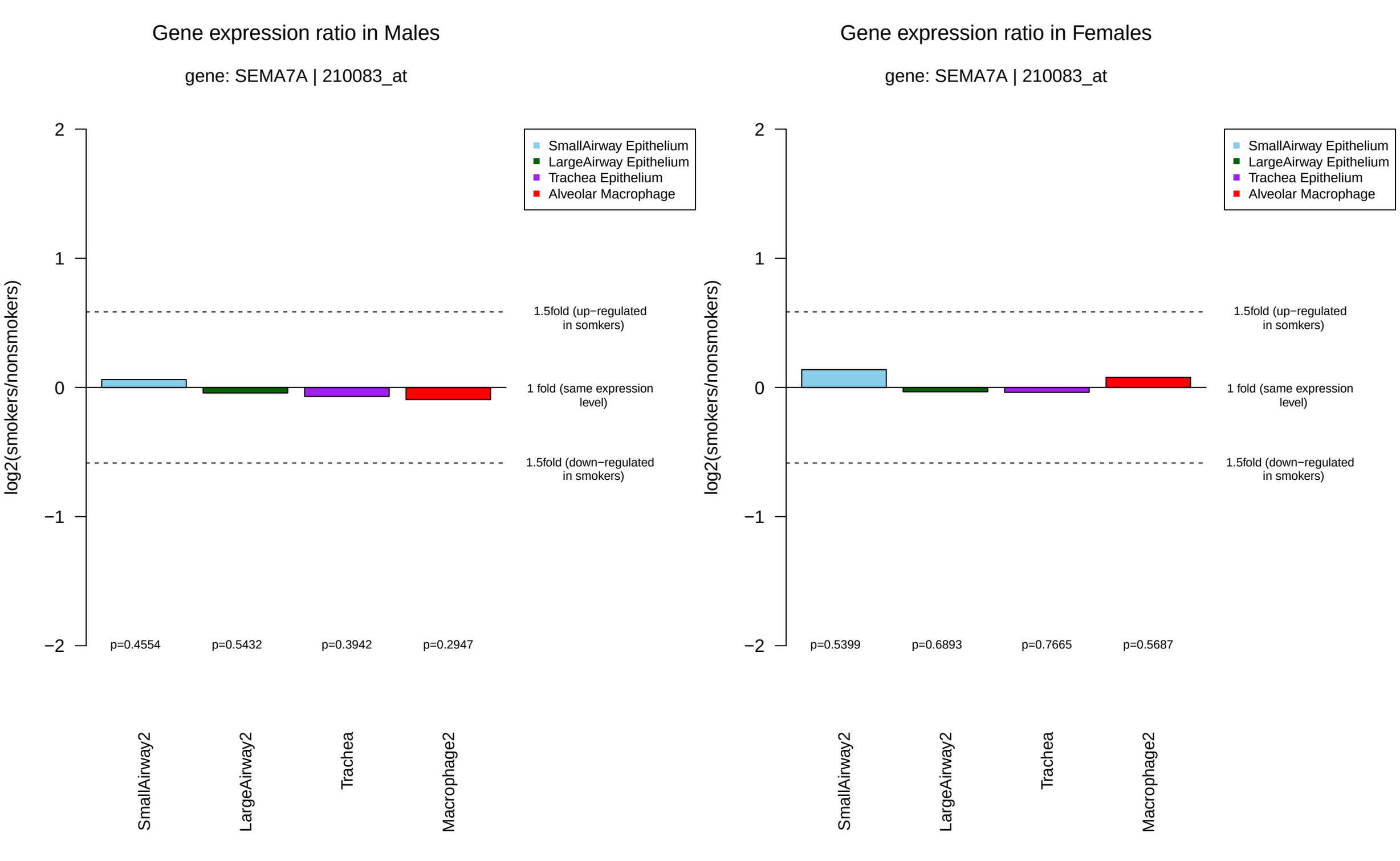
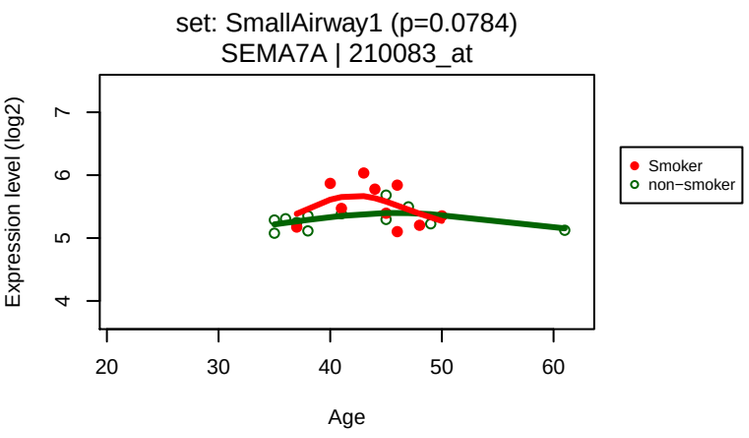
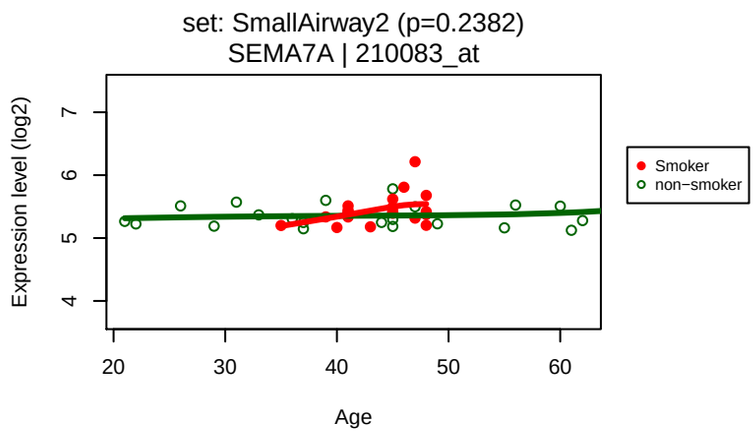
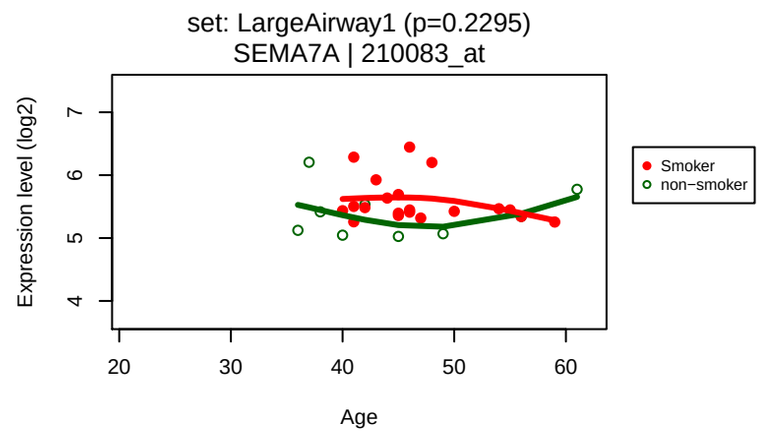

Gene expression of smokers and nonsmokers
gene: SEMA7A | 210083_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0784
p=0.2382
p=0.1188
p=0.2295
p=0.4238
p=0.2436
p=0.4854
p=0.6049
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

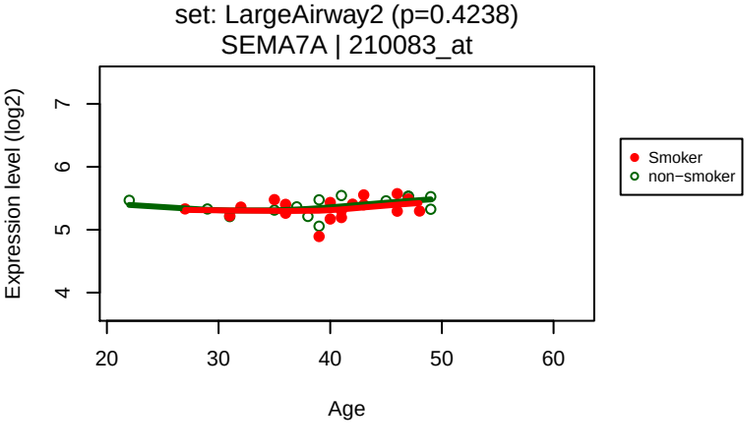

Gene expression ratio (smokers/nonsmokers)
gene: SEMA7A | 210083_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0784
p=0.2382
p=0.1188
p=0.2295
p=0.4238
p=0.2436
p=0.4854
p=0.6049
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

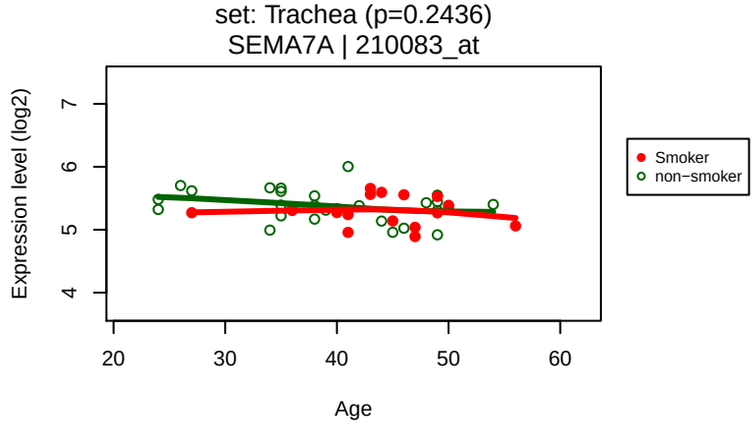

Gene expression ratio in Males
gene: SEMA7A | 210083_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4554
p=0.5432
p=0.3942
p=0.2947
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SEMA7A | 210083_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5399
p=0.6893
p=0.7665
p=0.5687
SmallAirway2
LargeAirway2
Trachea
Macrophage2

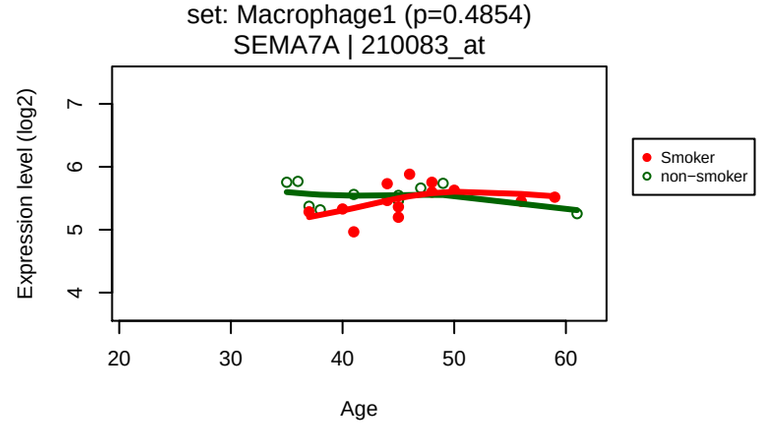

set: SmallAirway1 (p=0.0784)
SEMA7A | 210083_at
Expression level (log2)
Age
Smoker
non−smoker

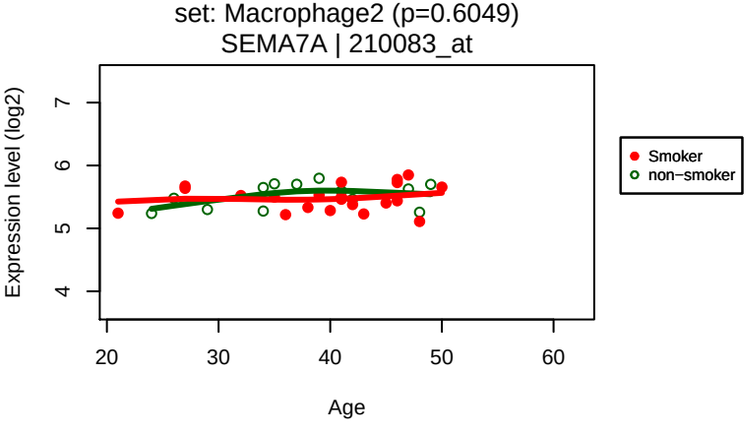

set: SmallAirway2 (p=0.2382)
SEMA7A | 210083_at
Expression level (log2)
Age
Smoker
non−smoker

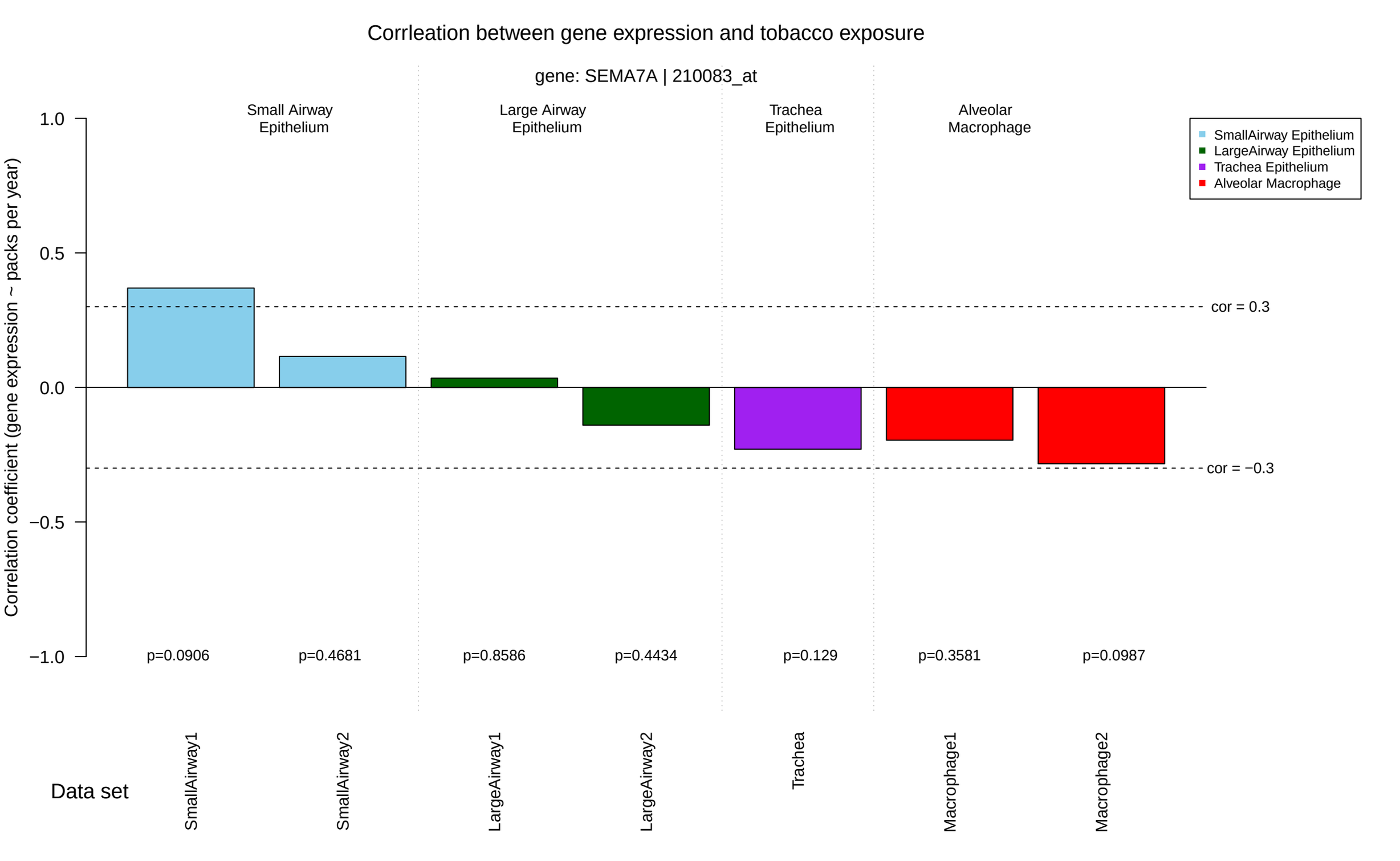

set: LargeAirway1 (p=0.2295)
SEMA7A | 210083_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4238)
SEMA7A | 210083_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2436)
SEMA7A | 210083_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4854)
SEMA7A | 210083_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6049)
SEMA7A | 210083_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SEMA7A | 210083_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0906
p=0.4681
p=0.8586
p=0.4434
p=0.129
p=0.3581
p=0.0987
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2